TrustSense-BMS: A Physics-Informed AI and Blockchain Integrated Battery Management System

By Team,

1. Yash Borade
2. Medha Karadiguddi
3. Sai Subhransu Dash
4. Krushna Sonar

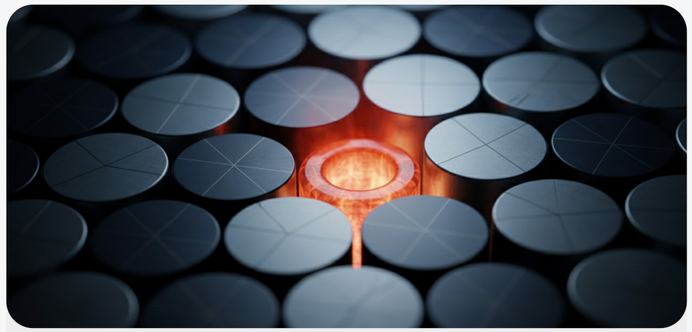
Agenda

1. Challenges and Current Limitations
2. TrustSense System Overview and Components
3. Outcomes and Future Directions

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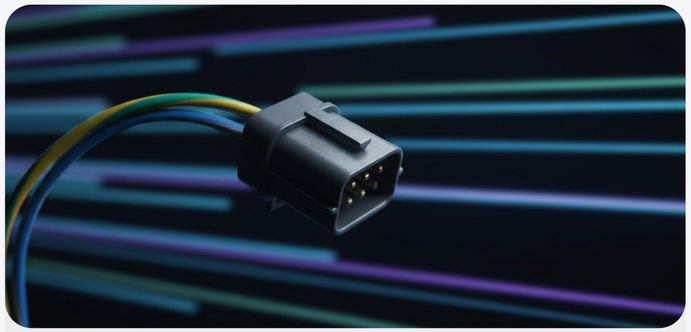
Challenges and Current Limitations

Why Trust Breaks



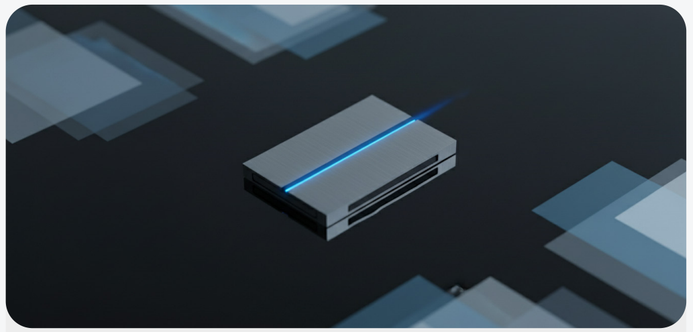
Thermal risk without full visibility

- A single undetected hot cell can trigger pack-level thermal runaway.
- Dense per-cell sensing is costly and adds weight and failure points.



In-vehicle data can be faked

- CAN bus traffic is typically unencrypted and implicitly trusted by controllers.
- Spoofed sensor messages can hide overheating or create false safe readings.



Battery history lacks tamper-proof evidence

- Mutable databases allow post-incident edits, weakening root-cause analysis.
- Untrusted lifecycle logs reduce compliance confidence and second-life resale value.

Where Current BMS Fails

Hardware-heavy thermal monitoring

- Many BMS designs rely on one sensor per cell or pack-level averaging.
- This increases wiring, weight, and cost without guaranteeing hotspot coverage.

Single-point blindness

- If a cell sensor fails, that cell can become effectively unobservable.
- A hidden hotspot can escalate before the controller reacts in time.

Implicit trust in CAN data

- Controllers often accept in-vehicle network messages without plausibility checks.
- Spoofed values can look normal digitally while violating real battery physics.

Weak accountability after incidents

- Mutable telemetry records make it hard to prove what happened and when.
- Outcome: reduced safety margin and limited forensic confidence for regulators.



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TrustSense System Overview and Components

The TrustSense Approach

Design principle

- Shift safety from hardware saturation to software-verified trust signals.

Pillar 1: AI virtual sensing

- Infer cell temperatures using physics and pack topology, not extra sensors.
- Restores heat visibility when a sensor fails or is missing.

Pillar 2: Cyber IDS

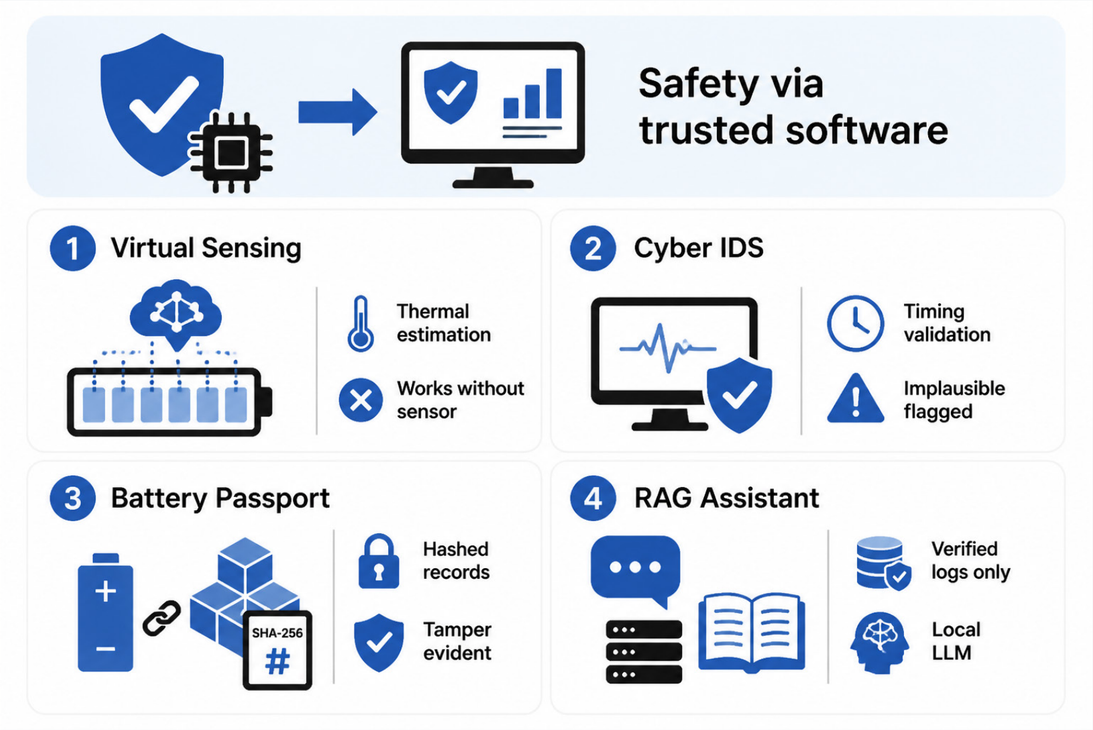
- Validate CAN telemetry by checking timing patterns and physical plausibility.
- Flags **thermodynamically impossible** values as likely spoofed messages.

Pillar 3: Battery Passport

- Anchor lifecycle events with **SHA-256 hashes** on Hyperledger Fabric.
- Creates tamper-evident provenance for incidents, compliance, and resale.

Pillar 4: RAG assistant

- A local LLM answers engineering questions from verified logs only.



How Data Flows

1) Telemetry ingestion

Pack voltage, current, and limited cell sensors stream into TrustSense.

2) AI virtual sensing

Physics-informed models infer per-cell temperature for full-pack visibility.

3) Real-time cyber screening

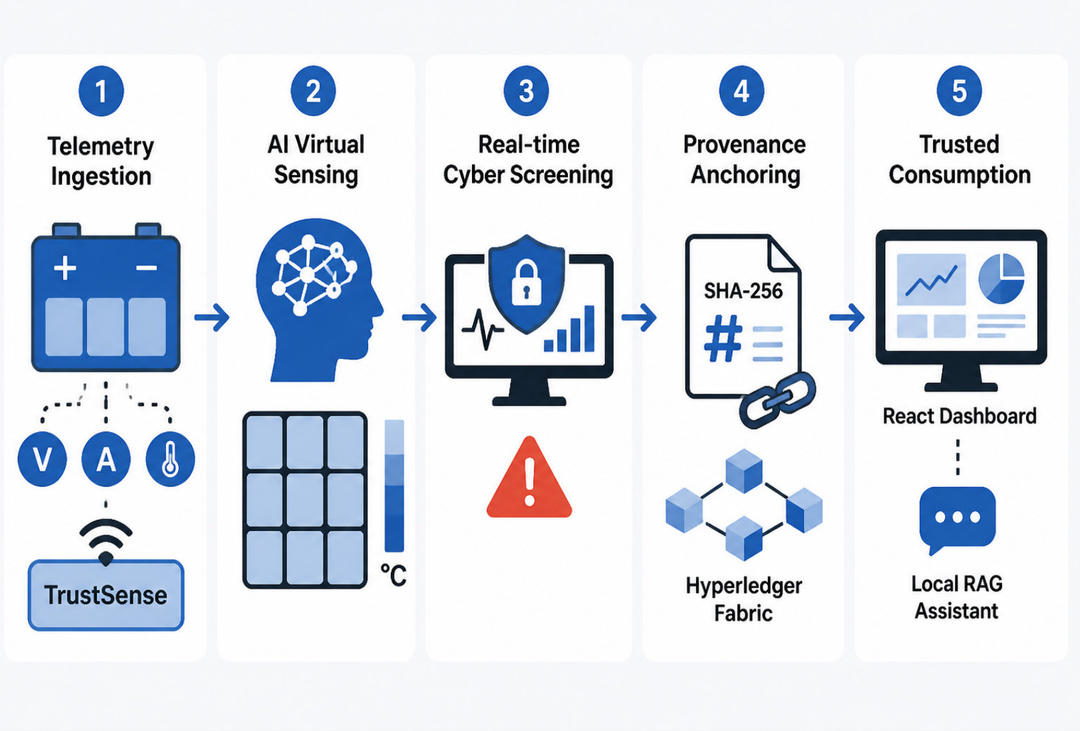
IDS checks timing and values; flags thermodynamically impossible changes.

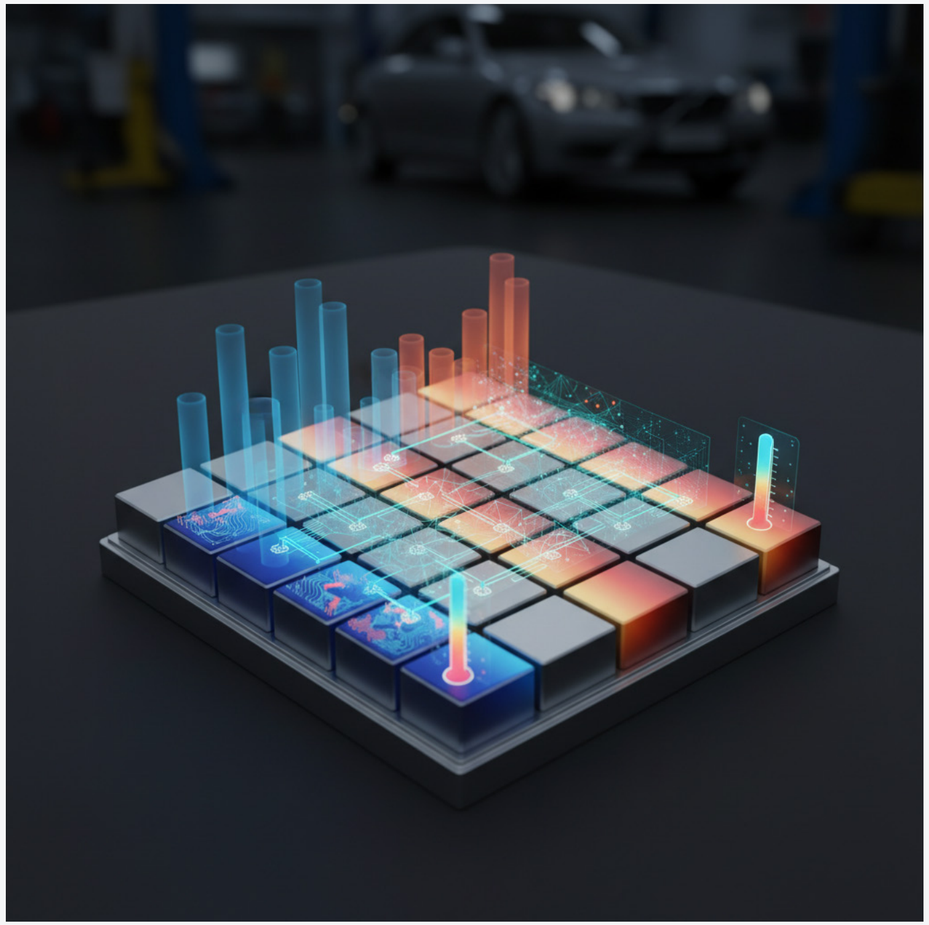
4) Provenance anchoring

Verified logs are SHA-256 hashed and committed to Hyperledger Fabric.

5) Trusted consumption

React dashboard and local RAG assistant read only verified records.





Virtual Sensing Engine

Physics-informed core

- PyBaMM models cell electrochemistry and thermal dynamics under load.
- This anchors predictions to known behavior, not just data patterns.

3D heat propagation model

- A Graph Neural Network learns heat transfer between neighboring cells.
- Pack layout becomes a graph, preserving adjacency and thermal coupling.

Virtual temperature sensing

- Infers per-cell temperatures where no physical sensor is installed.
- Target: reduce sensors by **>50%** while keeping RMSE **<1.0°C**.

Engineering impact

- Lower harness complexity, cost, and weight without losing hotspot visibility.

Cyber IDS Logic

Real-time CAN traffic modeling

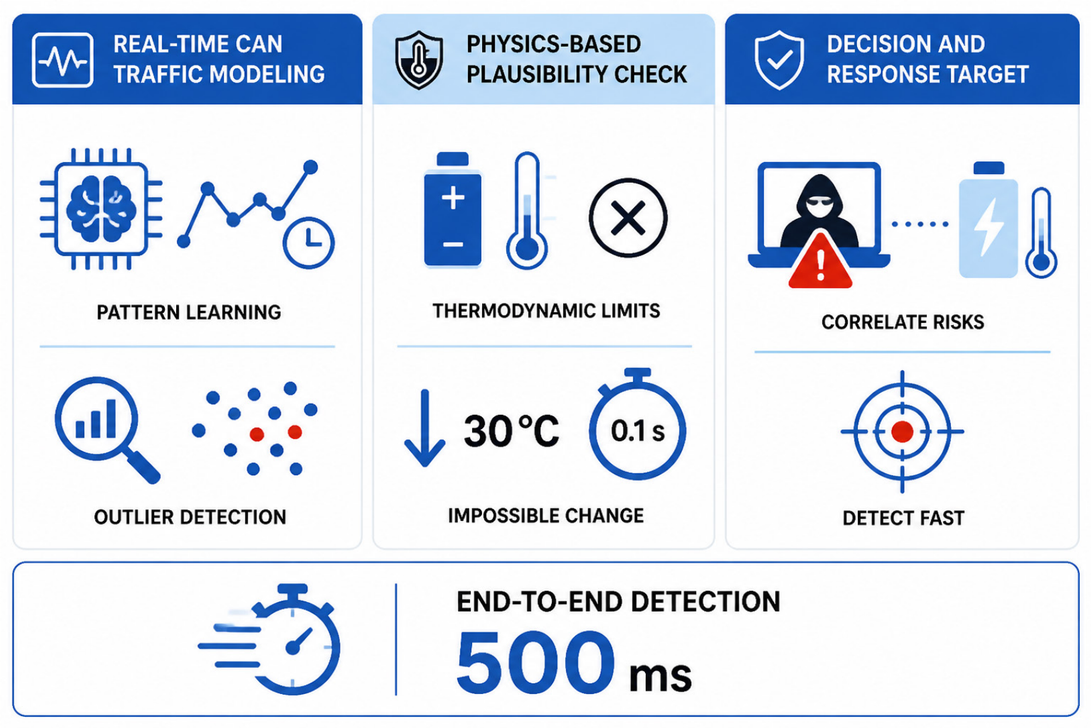
- Time-series Transformers learn normal message timing and sequence patterns.
- Isolation Forests flag outliers in values, cadence, and bus behavior.

Physics-based plausibility check

- Telemetry is validated against thermodynamic limits, not just signatures.
- Example: a 30 degrees Celsius drop in 0.1 seconds is impossible.

Decision and response target

- Correlate cyber anomalies with physics violations to reduce false alarms.
- Goal: detect spoofing and intrusion within **500 ms** end-to-end.



Battery Passport Layer

Off-chain data, on-chain proof

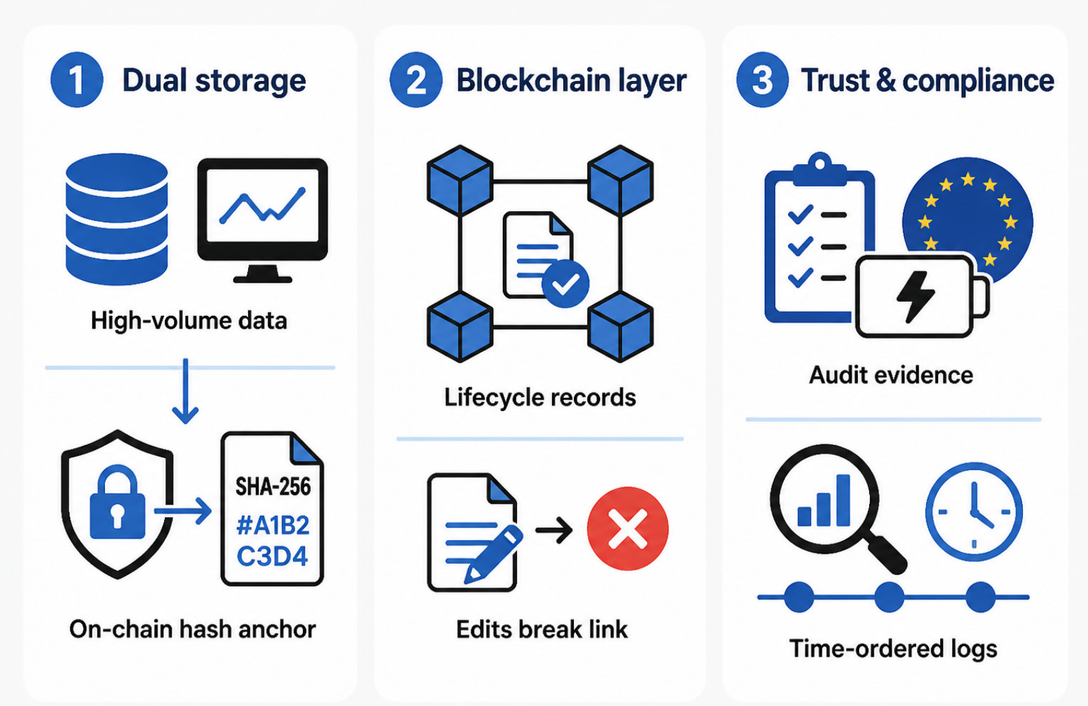
- Store high-volume telemetry off-chain to keep logging fast.
- Anchor each record with an on-chain **SHA-256 hash**.

Hyperledger Fabric passport

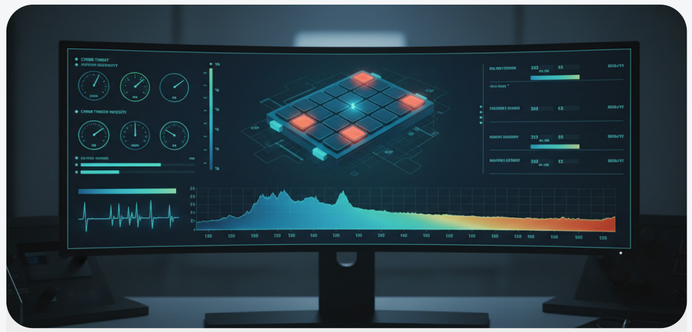
- Use Hyperledger Fabric to create traceable lifecycle events.
- Any later edit breaks the hash link and shows tampering.

Compliance and investigations

- Provides audit-ready evidence for the **EU Battery Passport** era.
- Improves post-incident forensics with verifiable, time-ordered logs.



Engineering Interface



Real-time engineering dashboard

- Shows pack and cell thermal status with clear, actionable views.
- Visualizes heat trends and hotspots to speed operator decisions.
- Displays cyber threat level alongside battery signals for context.



Local RAG assistant on trusted logs

- Engineers query a private Llama 3 assistant for incident triage.
- Answers are grounded only in verified blockchain-linked telemetry.



Faster investigations, less exposure

- Shortens root-cause analysis by unifying data, alerts, and history.
- Keeps sensitive operational data on-prem, not in external services.

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Outcomes and Future Directions

What Success Looks Like

Safety outcomes

- Virtual sensing exposes cell hot spots even with fewer sensors.
- Earlier anomaly visibility reduces overheating risk and fire escalation.

Cyber resilience

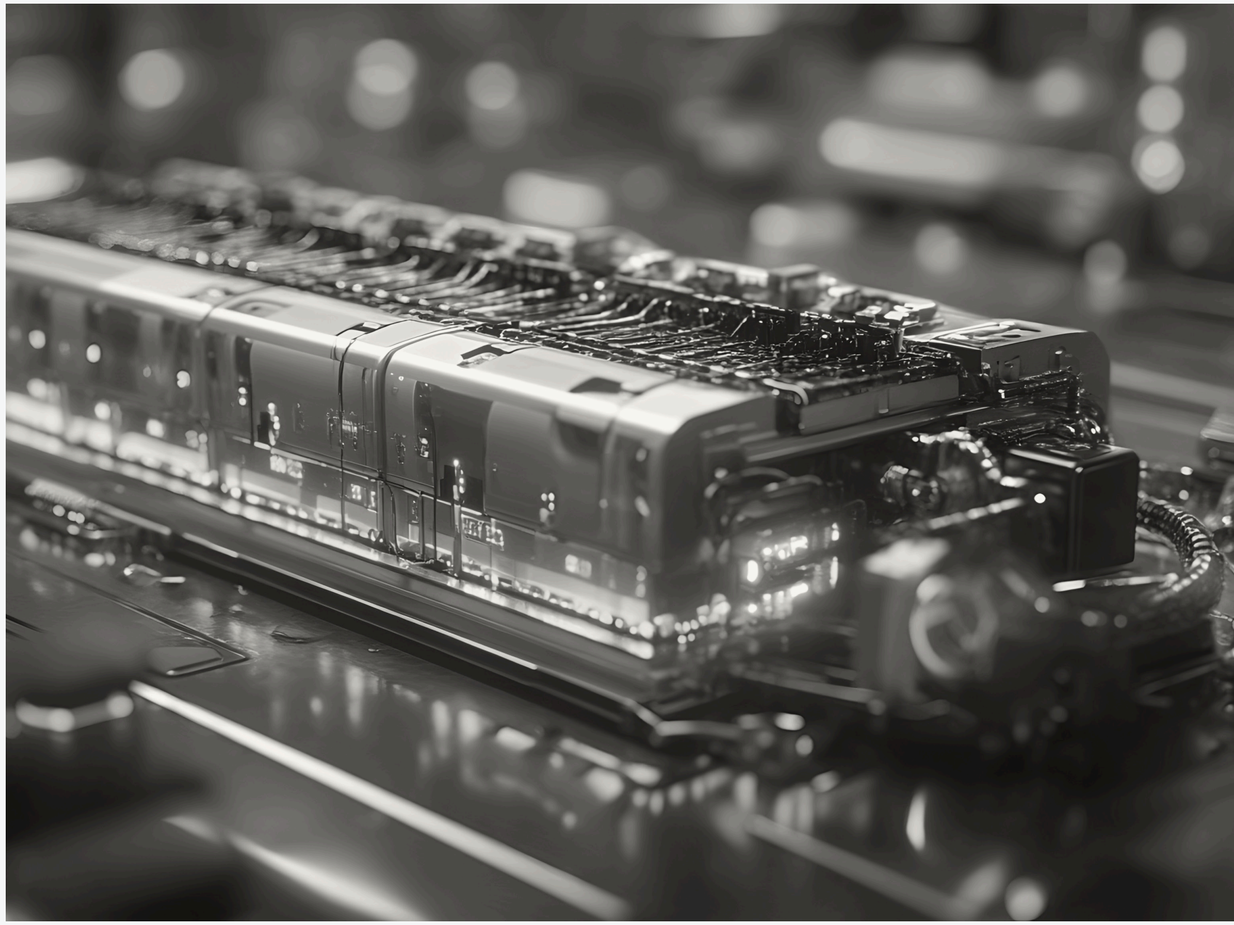
- IDS flags CAN messages that are statistically odd and physically impossible.
- Fast detection protects control decisions from spoofed battery telemetry.

Compliance and trust

- Immutable battery passport proves history was not tampered with.
- Audit-ready provenance supports recalls, resale confidence, and second life.

One deployable platform

- Unified dashboard turns trusted data into actionable operations signals.
- AI, cyber, and blockchain work together as one practical BMS layer.

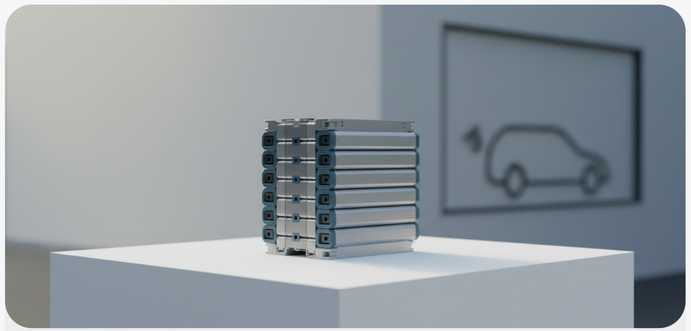


Next Steps



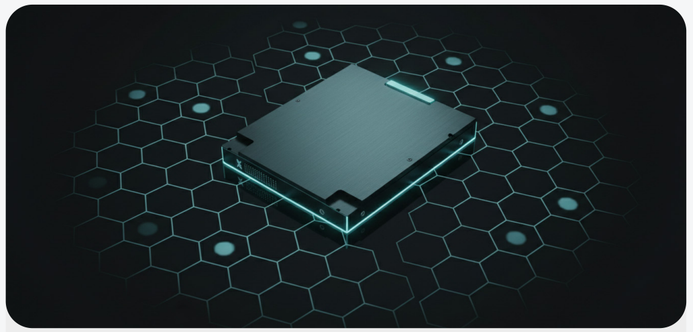
Edge deployment in vehicles

- Port virtual sensing and IDS models to Raspberry Pi or NVIDIA Jetson.
- Run low-latency inference onboard for faster, offline-safe decisions.



Second-life and V2G readiness

- Extend the Battery Passport schema for reuse, resale, and recycling.
- Add lifecycle events to support V2G participation and audits.



Scale trust across the ecosystem

- Standardize hashed logging so records stay tamper-evident end-to-end.
- Outcome: safer packs, lower hardware burden, and **provable accountability**.

Thank you